

EVT Tails on a Neural Quantile Body: Calibrated Multi-Horizon Return Distributions and the Price of Crash Insurance*

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Abstract

We train a single horizon-conditioned implicit quantile network (IQN) that outputs the full term structure of conditional return distributions—horizons of one week to three months—for roughly 543 liquid US single names over 2005–2026. Out of sample, under an annual walk-forward protocol with a 95-day purge and embargo, the distribution body and upper tail are essentially exactly calibrated at every horizon (empirical coverage 0.486–0.492 at $\tau = 0.50$, 0.693–0.699 at $\tau = 0.75$, 0.898–0.900 at $\tau = 0.95$), while the left tail is roughly twice too narrow: the 5% quantile is breached about 10% of the time. Heavier left-tail quantile sampling does not repair it (breach rate 0.108 versus 0.103), indicating an informational rather than architectural limit: crash-sized moves are driven by discrete idiosyncratic events not recoverable from price-based conditioning information. Grafting a generalized Pareto tail onto the neural quantile body, estimated strictly on past data within the walk-forward, restores calibration (5% coverage 0.056–0.063 and 1% coverage 0.012–0.014 at every horizon). We organize the method as GRAFT-Q (Generative Regression with Adaptive-Fringe Tails and conformal Quantiles): a horizon-embedded neural quantile surface, a generalized Pareto tail graft with splice-continuity constraints, and a pre-registered conformal layer exploiting cross-sectional exchangeability across the ~ 543 names at each date; every design choice is justified explicitly against the generative Bayesian computation (GBC) baseline of Polson and Sokolov (Section 4.6). The paper’s contribution is this calibrated *physical* return tail — sharper out of sample than the GARCH and filtered-historical-simulation forecasts that are the practitioner standard — and one economic result that follows directly from having it. Using the calibrated tail as the physical ($\mathbb{P}$) benchmark yields a conditional sharpening of the variance-risk-premium literature: even when the model correctly identifies elevated crash risk (its fat-tail trigger fires in 525 months against 6 for a GJR-GARCH- t benchmark), buying 5%-quantile puts loses 1.7% per month ($t = -1.9$). We read this not as mispricing but

*Working paper, first draft July 2026. Draft v2, July 2026 — revised scope. Comments welcome. All results are from walk-forward experiments described in Section 5; a live out-of-sample holdout beginning July 2026 is running at the time of writing. Economic results in Appendix A use a placeholder transaction-cost assumption (10% of premium) pending integration of measured OptionMetrics spreads; see Section 9 and Appendix A.5. A companion paper develops the residual-hybrid risk engine, the misspecification frontier, and generative-Bayesian extensions on broader universes; the two papers cross-reference and do not duplicate claims.

as evidence that the deep-put premium is a genuine *risk* premium: it survives conditioning on a correct physical forecast, which relocates the puzzle from forecasting to risk-bearing capacity, consistent with demand-based option pricing and probability-weighting demand. Banks calibrate risk-neutral surfaces to hedge, not to forecast $\mathbb{P}$; the persistent $\mathbb{P}$ - $\mathbb{Q}$ gap at the smile’s deep put wing is a premium natural sellers warehouse, not an arbitrage a better forecaster removes. Accordingly the model’s economic value is in the *tail, not the mean*: a model-driven GARCH $\times$ IQN regime switcher (Appendix A) manufactures no buy-side alpha but converts a crash-exposed short-premium book into a survivable one — the best crisis profile of any variant tested (net Sharpe 1.1–1.16 under a placeholder cost assumption, with measured execution costs and delisting tails pre-registered). The trading system is a *measurement* of what warehousing the premium currently pays, not a claim of inefficiency; pure variance-risk-premium harvest strategies without model input are reported separately.

Keywords: implicit quantile networks; extreme value theory; density forecasting; calibration; variance risk premium; crash insurance; walk-forward validation.

1 Introduction

The volatility surface embeds a forecast that is not meant to be one. When a bank calibrates an SVI, SABR, local-volatility, or Heston surface (????), the object being fit is the risk-neutral measure $\mathbb{Q}$: the surface prices hedges consistently and rules out static arbitrage, and its parameters are chosen so that the desk’s book replicates. Nothing in that exercise requires the surface’s implied probabilities to match the physical measure $\mathbb{P}$ under which returns are actually drawn. The wedge between the two—largest and most persistent at the deep out-of-the-money put wing—is the variance and crash risk premium documented across two decades of literature (?????). Measuring that wedge name-by-name, horizon-by-horizon, requires something the option-pricing toolkit does not supply: a calibrated estimate of the *physical* conditional return distribution, including its tails, at the same tenors the surface quotes.

This paper builds that estimate and stress-tests it. We adapt the implicit quantile network of ?—a method developed for distributional reinforcement learning in simulator environments—to the panel of US single-name equity returns. A single network, conditioned on a state vector, a quantile level τ , and a horizon h , outputs the entire term structure of conditional return quantiles from one week to three months. The estimation target is distributional calibration, verified out of sample under a walk-forward protocol whose purge and embargo rules we treat as first-class methodology rather than a robustness footnote. We call the assembled method *GRAFT-Q*: Generative Regression with Adaptive-Fringe Tails and conformal Quantiles. Its three stages are a horizon-embedded neural quantile surface (estimated), a generalized Pareto tail graft with splice-continuity constraints (estimated), and a conformal calibration layer built on cross-sectional exchangeability (specified and pre-registered; Section 4.5). The closest methodological antecedent outside finance is the generative Bayesian computation (GBC) program of ? and its quantile-based successor ?, which learn posterior and predictive quantile maps from simulated draws; Section 4.6 justifies, choice by choice, where and why financial data force departures from that baseline. The paper’s primary contribution is this calibrated physical distribution and its tail, benchmarked against the GARCH/filtered-historical-simulation standard; the crash-premium economics of Section 8 is a corollary of *having* that calibrated tail, not the reverse, and we make no claim of market inefficiency.

Three findings organize the paper. First, the approach works where the data can support it: across all horizons, the body and upper tail of the predicted distributions are calibrated essentially exactly out of sample. Second, it fails in a specific and instructive way: the left tail is roughly twice too narrow, and the failure is *informational*, not architectural—no amount of loss reweighting fixes it, because crash-sized moves in single names are driven by discrete idiosyncratic events (fraud revelations, regulatory actions, guidance shocks) that are not functions of the price-based conditioning set. The remedy that respects this diagnosis is not a bigger network but a statistical graft: a generalized Pareto tail (???) estimated on past exceedances and spliced onto the neural quantile body inside the walk-forward. Calibration is then restored at both the 5% and 1% levels. Third, with a calibrated physical tail in hand the economic question can be asked cleanly, and the answer is a conditional sharpening of the variance-risk-premium literature: the

crash premium survives a correct physical forecast. The model flags genuinely elevated tail risk far more often than a GARCH benchmark (525 months versus 6), yet buying puts at the 5% physical quantile in exactly those months loses 1.7% per month ($t = -1.9$). We read this as evidence that the premium is real compensation for bearing crash risk — not a mispricing a better forecaster could arbitrage away — which relocates the puzzle from forecasting to risk-bearing capacity. This is a corollary of the calibration result, not a competing claim.

1.1 Contributions

The paper makes four contributions. (i) *Methodological*: GRAFT-Q—a single horizon-conditioned IQN producing the full term structure of conditional return distributions, a walk-forward EVT splice that repairs the left tail using only past data, and a pre-registered cross-sectional conformal layer—with every design choice justified against the GBC baseline (??) in Section 4.6, together with a validation protocol—annual retraining, 95-day purge and embargo, point-in-time features, frozen decision rules, fresh-universe replication, and a live post-sample holdout—designed so that every reported number is attainable by an investor operating in real time. (ii) *Empirical*: out-of-sample calibration evidence at five quantile levels and all horizons, including the diagnosis that left-tail miscalibration in single names is an informational ceiling. (iii) *Economic (a corollary of the calibration)*: using the calibrated physical tail as the $\mathbb{P}$ benchmark against the risk-neutral surface, the deep-put crash premium is shown to survive conditioning on a correct physical forecast — evidence of a genuine risk premium rather than mispricing — tying the persistence of the $\mathbb{P}$ – $\mathbb{Q}$ wedge to demand-based option pricing (?) and probability weighting (??); the model’s trading value is risk-adjusted (tail control), not buy-side alpha. (iv) *Negative results as findings*: the model-timed crash-insurance dead end—plausible ex ante and documented with the same care as the successes (Section 8 and Appendix A.4); further strategy-design dead ends for designs without model input are reported separately with the non-model exhibits.

1.2 From physical simulators to financial markets

Distributional quantile methods of the kind we borrow were developed in domains—Atari emulators, physics simulators, game engines—whose statistical character is almost maximally unlike a financial panel (???). The generative Bayesian computation program (?) and its quantile successor (?) are the purest expression of the simulator-native design: they presuppose a forward model that can be simulated on the order of 10^6 times. Taking the transfer seriously, rather than treating the architecture as a plug-in, is what motivates most of this paper’s design choices. We identify six contrasts; each maps to a concrete feature of the project.

(1) *No data-generating law*. A simulator is a fixed, re-runnable data-generating process: validation consists of drawing fresh rollouts from the same law, and any quantity of interest can be estimated to arbitrary precision. Markets offer a single, non-repeatable realized path and no oracle. There is no “fresh rollout”; the closest admissible substitute is strict temporal separation between what the model sees and what it is scored on. This is why walk-forward estimation

with a purge and embargo (Section 5) replaces simulation-based validation, and why we regard the protocol as part of the method rather than as an evaluation detail.

(2) *Non-stationarity and regime breaks.* Simulator dynamics are stationary by construction. Market parameters drift and structures break outright—2008, 2020, the 2022 rate repricing, and the March-2026 correlation shock all sit inside our sample. Accordingly, all conditioning features are standardized on lagged windows, regime indicators are expanding-window percentiles computed point-in-time, the network is retrained annually so that its weights track the data-generating process at roughly the timescale on which that process moves, and the trading layer uses regime gates rather than fixed thresholds.

(3) *Reflexivity and adversarial adaptation.* Physics does not arbitrage itself: a projectile’s trajectory is indifferent to being predicted. Prices are not. Forecasts that are traded on alter the phenomenon being forecast (???), and documented predictability decays after discovery and publication (?). The compression of short-volatility premia after 2010 is the domestic instance of this. Three design consequences follow: edges must be assumed to decay, so negative results carry as much information as positive ones; point estimates of strategy performance are meaningful only alongside the process that generated them; and the only fully clean test is a live holdout, which we run from July 2026.

(4) *Tails driven by discrete idiosyncratic events.* In a simulator, extreme outcomes are draws from the same law that generates typical ones, and a sufficiently flexible density model can learn both. Single-name crash returns are dominated by events—fraud, FDA decisions, guidance withdrawals—whose arrival is largely invisible to price-based conditioning information. Section 7 presents the evidence: reweighting the quantile loss toward the left tail leaves the breach rate essentially unchanged (0.108 versus 0.103), which is what an informational ceiling looks like and what an architectural deficiency does not. The right response is not more architecture but a different estimator for the region the features cannot see: an EVT tail estimated from past exceedances, which assumes only that the *frequency and shape* of unforecastable extremes are stable, not that their timing is predictable.

(5) *Weak factors in high dimension versus strong laws in low dimension.* Simulator states are low-dimensional and dynamics are strong laws; financial conditioning information is a large collection of weak, correlated, drifting signals with very low signal-to-noise. Two of our findings are direct corollaries. Dimension reduction buys nothing at this data scale—raw 17 features, PCA-6, and block-PCA-6 are statistically indistinguishable (pinball loss—the tilted absolute-error score for a τ -quantile forecast, penalizing under-prediction with weight τ and over-prediction with weight $1 - \tau$, minimized by the true quantile (Section 4.1)—of 0.4217, 0.4226, 0.4231)—because with $\sim 450,000$ panel samples the network’s implicit regularization already handles 17 weak features. And the achievable goal is *calibration*, not point accuracy: no feature set will locate next month’s return, but a well-designed model can know the shape of its own uncertainty, which for option pricing is precisely the object of interest.

(6) *Endogenous observation.* A simulator’s observability is a design choice; a market’s is an economic outcome. Options coverage in our data grows from roughly 100 to 500 names over

the sample because listing, liquidity, and vendor coverage respond to the same economic forces that drive returns. Data availability is therefore correlated with the phenomenon under study, and survivorship and coverage bias must be modeled rather than assumed away. Section 3 describes our handling and its current limits (the CRSP delisting-return merge is pending and pre-registered as Appendix A.6).

The theme uniting the six contrasts is that in markets, the validation protocol, the treatment of tails, and the economics of the forecast’s own deployment are not implementation details—they are where the intellectual content lives. The method extensions in this paper are, in that sense, necessary rather than incidental.

2 Related literature

2.1 Distributional and quantile deep learning

Quantile regression originates with ?. Its deep-learning revival came, unusually, from reinforcement learning: ? argued for learning full return distributions rather than expectations; ? parameterized fixed quantile grids; and ? introduced the implicit quantile network, which samples τ at train time and embeds it with a cosine basis, thereby learning the entire quantile function with a single set of weights. We transfer the IQN to supervised panel forecasting and extend it with horizon conditioning. In the asset-pricing literature, ? established that neural networks extract conditional expected-return signals from large feature panels; our object is instead the full conditional distribution, and our loss is the pinball loss, a strictly proper scoring device for quantiles (?). In parallel, the generative Bayesian computation literature (?) and its quantile-prediction successor (?) learn quantile maps directly, prior-and-likelihood free, from large simulated datasets; GRAFT-Q shares the quantile-map object but must replace every simulator-dependent ingredient, a mapping made explicit in Section 4.6.

2.2 Volatility forecasting: the GARCH and HAR families

The conditional-heteroskedasticity tradition runs from ? and ? through the asymmetric GJR extension (?) that we use, in Student- t form on 21-day windows, as a timing benchmark and as one leg of the model-agreement signal. The realized-volatility strand (?) and its long-memory workhorse, the HAR model of ?, motivate our realized-vol feature block: the base features include realized volatility measured over cascaded windows precisely because HAR shows this parsimonious structure captures volatility’s memory.

2.3 Extreme value theory in finance

The Pickands–Balkema–de Haan theorem (??) justifies approximating exceedances over a high threshold by a generalized Pareto distribution; ? is the standard treatment, and ? pioneered the two-stage design—an econometric body with an EVT tail—that we adapt. Our splice differs in that the body is a neural conditional quantile function rather than a GARCH filter, and threshold

exceedances are defined in the model’s own probability coordinates, so the GPD corrects exactly the region where the network is informationally blind. Coverage is assessed with unconditional and interval-forecast tests in the spirit of ? (the Kupiec test: are there the right number of breaches?) and ? (the Christoffersen test: are breaches independent, not clustered in time?), and density calibration in the sense of ?. The generalized Pareto graft developed here is applied selectively in the companion paper’s multi-asset setting: on several volatility-index series the raw quantile body does not under-cover and the graft degrades accuracy, so the splice should be gated on measured tail under-coverage rather than applied universally.

2.4 The variance risk premium and option returns

That volatility trades rich under $\mathbb{Q}$ relative to $\mathbb{P}$ is among the most robust facts in derivatives: index and single-name variance swaps carry strongly negative premia (?), delta-hedged option positions lose systematically (?), put returns are anomalously negative (??), and a large share of the equity and variance premium is compensation for jump-tail fear (?). Our contribution to this literature is conditional: we show the premium survives even when the buyer of insurance conditions on a correct, calibrated physical-measure tail forecast.

2.5 Demand-based option pricing and probability weighting

? show that when dealers cannot perfectly hedge, end-user demand pressure moves option prices in proportion to unhedgeable risk—exactly the mechanism that lets the deep put wing stay rich: natural sellers must warehouse crash risk, and warehousing is expensive. On the demand side, cumulative prospect theory’s probability weighting (?) makes small-probability losses loom large, and ? show such preferences price skewness in equilibrium. Together these supply and demand distortions rationalize our headline negative result.

2.6 Evaluating density forecasts

The evaluation toolkit we adopt—in use or pre-registered (Section 5.4)—is standard and deliberately exhaustive. The pinball loss is consistent for quantiles among proper scoring rules (?); probability-integral-transform (PIT) diagnostics — forecasts are calibrated if and only if the transformed outcomes look uniformly distributed — assess full-density calibration (?); unconditional and conditional coverage tests are due to ? and ?; expected-shortfall backtests follow ?; pairwise forecast comparison uses ? (a t -test on the difference in loss between two forecasts); set-valued model comparison uses the model confidence set (the set of models statistically indistinguishable from the best one) of ?; and the economic value of distributional forecasts is assessed in the tradition of ?, here through the trading application of Appendix A.

2.7 Forecast decay and adaptive markets

? formalize why exploitable forecasting patterns self-destruct; ?? frame efficiency as an evolutionary, regime-dependent property; ? measure post-publication decay directly; and ? document

how fragile in-sample predictability is out of sample. This literature underwrites Section 1.2’s argument and our insistence on walk-forward estimation, frozen rules, and a live holdout.

3 Data

3.1 Universe and panel

The universe comprises approximately 543 liquid US single-name equities with listed options, observed 2005–2026. Combining names, dates, and the four prediction horizons (one week, two weeks, one month, three months—settled on business-day grids) yields roughly 450,000 panel row $\times$ horizon training samples. Implied-volatility surfaces are from Bloomberg and are available at monthly tenors only; weekly-tenor analysis now travels with the separately reported non-model VRP-harvest exhibits (Appendix A.6). The options universe grows from roughly 100 names early in the sample to roughly 500 by its end, an endogeneity discussed below.

3.2 Features

The base specification uses nine features per name-date: a realized-volatility block (multiple lagged windows, in the HAR spirit), balance-sheet leverage, the name’s own implied-volatility level and skew, the VIX level, the volatility term-structure slope, and the futures basis. An extended 17-feature specification adds the VIX9D/VIX ratio, the 10-year Treasury yield, the 2s10s slope, high-yield and investment-grade OAS, the trailing 63-day market return, drawdown-from-peak, and distance-from-200-day-moving-average. All features are standardized using lagged windows only. Earnings-announcement dates have been collected for 81 names; a days-to-earnings feature is staged but *not* in any model reported here—we return to why it matters in Section 7.

3.3 Endogenous coverage and survivorship

Because options listing and vendor coverage respond to size, liquidity, and investor interest, the panel’s growth from ~ 100 to ~ 500 names is itself an economic outcome (contrast (6) of Section 1.2). Names enter the panel when covered and exit on delisting. Survivorship handling is partial in this draft: the panel is point-in-time with respect to entry, but the merge of CRSP delisting returns—which govern the true left tail of exiting names—is pending and is pre-registered, with the exhibits it will restate enumerated in Appendix C. Until that merge, single-name left-tail economics (though not the calibration statistics, which are computed on observed returns) should be read as potentially favorable to short-put strategies.

4 Method: GRAFT-Q

GRAFT-Q (Generative Regression with Adaptive-Fringe Tails and conformal Quantiles) has three stages: (i) a horizon-embedded neural quantile surface (Sections 4.1–4.2), (ii) a generalized

Pareto tail graft with splice-continuity constraints (Section 4.4), and (iii) a conformal calibration layer built on cross-sectional rather than temporal exchangeability (Section 4.5). Stages (i) and (ii) are estimated and produce every number in this paper; stage (iii) is fully specified and pre-registered, with results to follow in a revision. Section 4.6 justifies each design choice against the GBC baseline.

4.1 Stage 1: implicit quantile networks

Let $r_{i,t \rightarrow t+h}$ denote the return of name i over horizon h beginning at t , and $x_{i,t}$ the feature vector. We seek the conditional quantile function $Q(\tau \mid x, h) = F^{-1}(\tau \mid x, h)$ for all $\tau \in (0, 1)$ simultaneously. Following ?, a sampled quantile level $\tau \sim U(0, 1)$ is embedded through a cosine basis,

$$\phi_j(\tau) = \text{ReLU}\left(\sum_{k=0}^{n-1} \cos(\pi k \tau) w_{kj} + b_j\right), \quad (1)$$

and merged multiplicatively with the state embedding $\psi(x_{i,t}, h)$, so a single set of weights θ represents $\hat{Q}_\theta(\tau \mid x, h)$ over the whole unit interval. Training minimizes the pinball (quantile) loss

$$\mathcal{L}(\theta) = \mathbb{E}_{\tau, (i,t,h)} \rho_\tau(r_{i,t \rightarrow t+h} - \hat{Q}_\theta(\tau \mid x_{i,t}, h)), \quad \rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\}), \quad (2)$$

which is minimized in expectation exactly when $\hat{Q}_\theta$ is the true conditional quantile function (??). Relative to a fixed quantile grid (?), the implicit parameterization shares statistical strength across all of τ , enforces smoothness in the quantile direction, and lets us query arbitrary levels—including the deep quantiles the option smile quotes—without retraining.

In subsequent work on residual-space quantiles (companion paper), the combination of tail-aware τ -sampling (half uniform, half Beta(0.3, 0.3)), monotone rearrangement, and a split-conformal level shift repaired the neural tail’s under-coverage — consistent with this paper’s finding that heavier *sampling* alone (v2c) does not: the missing ingredients were the monotonicity constraint and the distribution-free recalibration, not more tail draws.

4.2 Stage 1, continued: horizon conditioning

Horizon h enters as a conditioning input alongside x , so a single network emits an internally consistent *term structure* of distributions rather than four independently estimated models. This mirrors the object on the other side of the wedge—the volatility surface is likewise one object indexed by tenor—and shares information across horizons: distributional shape at one month informs three months. The training sampler draws (i, t, h) uniformly over the panel and τ uniformly (with a heavier-left-tail variant evaluated in Section 6.2). Figure 1 sketches the architecture.

4.3 Ensembling

We train an ensemble of $K = 3$ networks differing in initialization and data-order seed and average their quantile outputs. The ensemble reduces seed-level variance in deep quantiles,

where a single network’s estimates are noisiest.

4.4 Stage 2: the walk-forward EVT tail graft

The Pickands–Balkema–de Haan theorem states that, for a wide class of distributions, exceedances over a high threshold converge to a generalized Pareto law. We apply it in the model’s own probability coordinates. Let $u_{i,t,h} = \hat{Q}(0.05 \mid x_{i,t}, h)$ be the neural body’s 5% quantile. Historical standardized exceedances below the model’s threshold—collected *only from dates preceding the current training cutoff*—are fit by maximum likelihood to a GPD with shape ξ and scale β , and the spliced left tail is

$$\hat{F}(r \mid x, h) = 0.05 \cdot \bar{G}_{\xi, \beta}(u_{i,t,h} - r), \quad r < u_{i,t,h}, \quad (3)$$

where $\bar{G}_{\xi, \beta}$ is the GPD survival function. The construction has two properties worth emphasizing. First, it is continuous at the splice point by construction and leaves the calibrated body ($\tau \geq 0.05$) untouched. Second, it is estimated inside the walk-forward: the GPD parameters used at date t never see data after t ’s training cutoff, so the splice inherits the protocol’s out-of-sample discipline. Conceptually this is ? with the GARCH filter replaced by a neural conditional quantile function.

4.5 Stage 3: a cross-sectional conformal layer (pre-registered)

Conformalized quantile regression (?) adjusts quantile-regression bands with a held-out calibration set to obtain finite-sample coverage guarantees under exchangeability (?). Time-series data violate temporal exchangeability in exactly the ways Section 1.2 catalogs, which is why naive conformal wrappers on financial forecasts inherit no guarantee. GRAFT-Q’s third stage exploits a different exchangeability direction: *cross-sectional* exchangeability across the ~ 543 names at a fixed date. Conditional on a date and horizon, the names’ standardized conformity scores are far closer to exchangeable than any collection of time points, because they share the date’s regime by construction. The layer computes, per date, CQR-style conformity scores across the cross-section and adjusts the spliced quantile surface so that finite-sample coverage holds on average across names at that date. This stage is specified and pre-registered but not yet reflected in any reported number; the numbers in Sections 6–8 come from stages (i)–(ii), whose two-decade walk-forward coverage—with breach-rate diagnostics in the tradition of ? and ?—is the non-exchangeable world’s substitute for a finite-sample theorem. The registered hypothesis is that the conformal layer tightens the residual 5% overcoverage (0.056–0.063) toward nominal without disturbing the body.

4.6 Design decisions relative to the GBC baseline

The GBC program (?) and its quantile-prediction form (?) are the natural baseline against which to justify GRAFT-Q’s departures, precisely because they represent the simulator-native ideal: quantile maps learned prior-and-likelihood free from a re-runnable forward model. The

Table 1: Design decisions relative to the GBC baseline (??).

Design axis	GBC baseline	GRAFT-Q choice and finance justification
Horizon treatment	One prediction task per trained map; horizon at best a generic input	Horizon embedding conditioning a single network across the whole term structure: the object of economic interest (the surface) is indexed by tenor, and cross-horizon information sharing stabilizes deep quantiles
τ -sampling scheme	Uniform τ against simulated draws	Uniform retained; heavy-left v2c rejected (0.108 vs. 0.103). A later companion-paper experiment shows the effective fix is tail-aware Beta(0.3,0.3) sampling <i>plus</i> monotone rearrangement and a conformal layer (raw 99% breach 3.2% $\rightarrow$ 0.92%), not heavier sampling alone
Monotonicity in τ	Crossing acknowledged, unenforced	Monotone-in- τ penalty on the body; the deep tail—where crossing concentrates—is replaced outright by the parametric graft
Tail treatment	Smoothness theory (Prop. 1) requires continuous Q'' , excluding fat tails	EVT/GPD graft with splice continuity: fat tails are the object of interest, and the GPD is the asymptotically justified form for exactly the region the smoothness assumptions exclude (??)
Validation	Fresh draws from the simulator; $\sim 10^6$ simulations presupposed	Walk-forward with 95-day purge/embargo: markets are one non-repeatable path with no oracle (Section 1.2, contrast 1)
Ensembling	Single trained generator	$K = 3$ ensemble: seed-level variance in deep quantiles is material when signal-to-noise is low
Finite-sample theory	Borrowed $O(N^{-4})$ approximation bound	Pre-registered conformal layer on <i>cross-sectional</i> exchangeability (Section 4.5): the exchangeability direction that survives in a panel
Data regime	Single simulable model, unlimited draws	Panel training across ~ 543 names and $\sim 450k$ samples: cross-sectional breadth substitutes for re-runability, and calibration—not point accuracy—is the attainable target

authors are candid about the frame’s limits, and each stated limit lands exactly where financial data live: their smoothness result (Proposition 1) requires a continuously twice-differentiable quantile function, excluding the fat-tailed case; quantile crossing (predicted quantiles coming out of order across τ — e.g. the 10th-percentile forecast landing above the 50th) is acknowledged but not enforced away; the available approximation theory is a borrowed $O(N^{-4})$ bound rather than a guarantee adapted to the estimator; and the construction presupposes a simulable forward model with on the order of 10^6 draws. Table 1 maps each GRAFT-Q design choice to its GBC counterpart and the finance-specific reason for departing.

5 Validation protocol

5.1 The thirty-years-on-thirty-years fallacy

A common design in the machine-learning-for-finance literature fits a model once on a long history and reports metrics on the same span, or on a single terminal split whose hyperparameters were nonetheless tuned against it. Call this the thirty-years-on-thirty-years fallacy. It fails twice over: the weights see the future (label leakage across overlapping horizons is endemic when targets span up to three months), and—more insidiously—the *researcher* sees the future, because every design decision made after glancing at a terminal-split result is a form of test-set reuse. Both channels inflate reported performance in ways that are invisible in the published number (??). Our protocol is built to close both.

5.2 Walk-forward with purge and embargo

The model predicting date t is trained only on data whose labels end at least 95 days before t . Since the longest horizon is roughly 63 business days, the 95-day embargo exceeds the longest horizon plus slack: no training label overlaps any prediction window, eliminating leakage through overlapping returns. Retraining is annual, so weights are refreshed at the timescale of regime drift. All signals are standardized on lagged windows; regime classifications are expanding-window percentiles computed point-in-time; and simulated entries settle over $t+1$ through $t+h$, so no trade uses information unavailable at its open. Figure 2 illustrates.

5.3 Design-decision overfitting

The subtler channel—the researcher as leak—is handled with three instruments. *Frozen rules*: strategy logic and thresholds were fixed before evaluation on the full panel; parameter selection anywhere in the pipeline uses nested walk-forward only, never the reporting sample. *Fresh-universe replication*: headline results were re-run on a 496-name universe assembled independently of the development universe; conclusions were unchanged. *Live holdout*: all rules were frozen as of July 2026, and performance on data arriving after that date constitutes a holdout that no design decision can have touched. At the time of writing this holdout is accruing; it will be reported in a revision, and we commit to reporting it regardless of outcome.

5.4 Pre-registered evaluation battery

To keep the evaluation itself from becoming a garden of forking paths, the full test battery is fixed here by name. *In use in this draft*: unconditional coverage (breach-rate) statistics in the tradition of ?, reported by quantile level and horizon (Table 3); mean pinball loss for model comparison (?). *Pre-registered, results to follow*: conditional coverage and independence tests of ?, stratified by volatility regime; probability-integral-transform histograms and their autocorrelation diagnostics (?); expected-shortfall backtests of the spliced tail via the Z -statistics of ?; pairwise ? tests of the IQN against the GJR-GARCH- t benchmark and of the dimension-reduction variants

Table 2: Same-rows audit: IQN vs unit-matched GJR-GARCH- t , average pinball loss on identical (ticker, date, horizon) rows.

Horizon (days)	IQN	GJR-GARCH- t	ratio	DM t
5	0.01302	0.01245	1.046	+12.3
10	0.01871	0.01769	1.058	+17.2
21	0.02762	0.02602	1.062	+18.1
42	0.03895	0.03710	1.050	+14.0
63	0.04757	0.04552	1.045	+12.3

The parametric model wins by 4.5–6.2% at every horizon, all strongly significant. The gap is concentrated in the tails (downside $\tau=.05 \approx 7\%$ worse at all horizons; the body $\tau=.50$ is nearly tied at 1.02–1.05). Applying the paper’s past-only pooled tail recalibration fixes coverage (5% breach 0.071–0.089 $\rightarrow$ 0.051–0.064) and roughly halves the pinball gap at long horizons (ratios $\rightarrow$ 1.029–1.048) — but GJR-GARCH- t still wins: the deficiency is not purely a fixable tail-calibration artifact.

against one another; a model confidence set (?) over the full variant tournament, for which—given the third-decimal pinball spreads of Section 6.5—we register the expectation that all three feature specifications survive; and an economic-value assessment in the spirit of ?, implemented through the trading application of Appendix A. Registering the battery before computing it is the evaluation-side analogue of the frozen-rule discipline of Section 5.3.

5.5 A unit-matched parametric audit (added July 2026)

An earlier internal comparison showed the network beating a t -GARCH benchmark by roughly 38%; an audit found the two models had been scored on different universes and possibly different target scalings, and that figure is retired. The corrected comparison scores both models on identical (ticker, date, horizon) rows — 102,391 rows, 111 names, horizons $h = 5/10/21/42/63$ trading days — against a unit-matched GJR-GARCH- t (zero-mean, Student- t innovations, annual walk-forward refits, h -day quantiles by Monte Carlo path simulation).

This audit sharpens rather than undermines the paper’s claims: on single-name daily-to-quarterly equity forecasting the leverage-GARCH is the honest benchmark and the network does not beat it. The network’s demonstrated advantages are exactly the ones GARCH structurally lacks — calibrated full-distribution term structures after the EVT graft, forecasting unseen assets, and the amortized cross-sectional problem — and the trading results of Appendix A never depended on beating GARCH on average loss.

Table 3: Out-of-sample calibration by quantile level. Entries are empirical coverage (fraction of realized returns at or below the predicted quantile); ranges span the four horizons (1w–3m). All quantities from walk-forward predictions, 2005–2026.

Quantile level	Nominal coverage	IQN body (pre-fix)	Heavy left-tail sampling (v2c)	IQN body + EVT splice
$\tau = 0.01$	0.010	—	—	0.012–0.014
$\tau = 0.05$	0.050	≈ 0.103	0.108	0.056–0.063
$\tau = 0.50$	0.500	0.486–0.492		0.486–0.492
$\tau = 0.75$	0.750	0.693–0.699		0.693–0.699
$\tau = 0.95$	0.950	0.898–0.900		0.898–0.900

Note: the EVT splice modifies only the region below the model’s 5% quantile; body and upper-tail rows are unchanged by construction. The v2c variant reweights the training distribution of τ toward the left tail; its failure to move the 5% breach rate (0.108 vs. 0.103) is the informational-ceiling diagnostic of Section 7.

6 Calibration results

6.1 Body and upper tail: essentially exact

Table 3 and Figure 3 report unconditional coverage of the walk-forward predicted quantiles against realized returns, by quantile level, with ranges across the four horizons. At $\tau = 0.50$ empirical coverage lies in 0.486–0.492 at every horizon; at $\tau = 0.75$, 0.693–0.699; at $\tau = 0.95$, 0.898–0.900. Deviations from nominal are a fraction of a percentage point across two decades of out-of-sample months—for practical purposes, the body and upper tail of the conditional distribution are exactly calibrated, at every horizon, from a single network. These are unconditional (Kupiec-style) statistics; the conditional-coverage, PIT, and expected-shortfall diagnostics of the pre-registered battery (Section 5.4) will accompany them in a revision.

6.2 The left tail: a two-times-too-narrow failure

The same model’s 5% quantile is breached roughly 10% of the time—the left tail is about twice too narrow. The natural architectural response is to make the training loss care more about the tail: variant v2c samples τ with substantially heavier weight on low quantile levels. It does not work: the breach rate moves from approximately 0.103 to 0.108, statistically nothing. When reweighting the objective toward a region fails to improve fit in that region, the binding constraint is not the objective but the information set. We develop this diagnosis in Section 7.

6.3 The EVT splice restores calibration

Grafting the walk-forward GPD tail (Section 4.4) onto the neural body brings the 5% breach rate to 0.056–0.063 and the 1% breach rate to 0.012–0.014, at every horizon. Residual deviations are small and conservative in direction relative to the raw body’s optimism. The splice achieves this without touching the exactly calibrated body, without any forward-looking data, and with two

estimated parameters—an instructive ratio of parameters to problem relative to the architectural alternatives.

6.4 A cautionary result: naive imputation destabilizes training

The 17-feature specification includes macro series with missing early-sample values. A variant that zero-imputed those NaNs destabilized training: predicted distributions degenerated, with coverage collapsing toward the median. The mechanism is mundane—zero is a meaningful value for standardized macro features, so imputation injected a spurious, regime-correlated signal—but the failure mode (silent degradation of *distributional* output while point metrics look survivable) is specific to density models and worth flagging for practitioners. Proper imputation is required; results reported here use the correctly imputed pipeline.

6.5 Dimension reduction buys nothing at this data scale

A tournament compared the raw 17-feature specification against PCA reduction to six factors and a block-structured PCA (six factors extracted within economically defined blocks). Mean pinball losses are 0.4217, 0.4226, and 0.4231, respectively (Figure 4)—raw features win by a hair, and nothing separates the three economically. With $\sim 450,000$ samples and 17 weak features, the network’s own regularization suffices; the folk wisdom that financial deep learning requires aggressive feature compression is, at this scale, not supported. This is contrast (5) of Section 1.2 made quantitative. Formal comparison—? pairs and a model confidence set (?) over the tournament—is pre-registered in Section 5.4, with the registered expectation that no specification is eliminated.

7 What the model cannot know

The left-tail evidence assembles into a statement about information rather than architecture. Three observations. First, the failure is asymmetric: the upper tail at $\tau = 0.95$ is calibrated to within twenty basis points of coverage while the lower 5% quantile is breached at twice its nominal rate. A capacity- or loss-driven failure would not know the sign of returns. Second, the failure is insensitive to the objective: heavier left-tail τ -sampling (v2c) moves the breach rate from 0.103 to 0.108—if the features contained the missing information, reweighting the loss is exactly the intervention that would extract it. Third, the events behind single-name crash returns are, on inspection, dominated by discrete idiosyncratic news: fraud revelations, FDA actions, guidance withdrawals, litigation. None of these are functions of realized volatility, implied volatility, leverage, or macro state—the price-based conditioning set is *informationally blind* to their timing.

We call this the *idiosyncratic-event ceiling*: a bound on achievable left-tail sharpness for any model conditioned on price-derived features. Its practical corollaries: (i) the correct fix is statistical, not architectural—the EVT splice models the frequency and shape of what cannot be timed, which is why it restores calibration where a better network could not; (ii) the ceiling can

be *raised* only by adding event information to the conditioning set—the staged days-to-earnings feature (81 names collected) is the first such addition, chosen because earnings dates are the one scheduled component of the event flow; and (iii) the ceiling disciplines interpretation of the economics: a model that cannot time idiosyncratic crashes can still price their frequency, and Section 8 shows that even that is enough to identify systematic mispricing—just not in the direction insurance buyers would hope.

8 The price of crash insurance

8.1 The experiment

If the market’s put wing overstates physical crash probabilities on average, perhaps it understates them conditionally—in months where tail risk is genuinely elevated, insurance might be cheap. A calibrated tail model is exactly the instrument to test this. We define a fat-tail trigger from the IQN’s predicted distributions and compare it with the analogous trigger from the GJR-GARCH- t benchmark. The two see different worlds: the IQN trigger fires in 525 months across the panel; the GARCH trigger fires in 6 (Figure 5a). Conditional on the IQN trigger, buying puts struck at the model’s 5% physical quantile loses 1.7% per month ($t = -1.9$) net (Figure 5b). Model-timed crash insurance still loses.

8.2 Interpretation: who closes the $\mathbb{P}$ – $\mathbb{Q}$ gap?

The result sharpens the variance-risk-premium (VRP: the premium option sellers earn on average because option-implied variance exceeds subsequently realized variance) literature’s unconditional findings (????) into a conditional statement: the crash premium survives conditioning on a correct physical-measure tail forecast ($\mathbb{P}$ versus $\mathbb{Q}$: the physical measure under which returns are actually drawn, as opposed to the risk-neutral measure embedded in option prices). ? attribute much of the premium to jump-tail fear; our evidence says the fear is priced richly even when a calibrated model agrees the tail is fat. Two mechanisms jointly rationalize persistence. On the demand side, probability weighting in cumulative prospect theory (?) inflates willingness to pay for small-probability protection, and ? show such preferences survive aggregation. On the supply side, ? show dealer risk-bearing constraints translate demand pressure into price: the marginal seller of a deep put cannot hedge the crash state and must warehouse it.

This is also where the paper’s framing closes. Banks calibrate $\mathbb{Q}$ —SVI, SABR, local volatility, Heston—to hedge and to rule out arbitrage against other options, not to forecast $\mathbb{P}$ (????). Nothing in the sell side’s objective function pushes the smile’s put wing toward physical probabilities. The natural counterparty who could close the gap—a systematic seller of overpriced crash insurance—must warehouse exactly the risk the dealers will not, and so demands a premium for it. The gap persists not because no one sees it but because seeing it is not the binding constraint; balance-sheet capacity for negative skew is. The trading system of Appendix A is best read as a measurement of what that warehousing is currently paid.

9 Limitations

We list the limitations we consider material, in rough order of how much they could move the economic (not the calibration) results.

Transaction costs are a placeholder. All net economic results assume costs of 10% of premium. This is an assumption, not a measurement; it is flagged in every affected exhibit, and Appendix A.5 specifies the measured-spread cost model (OptionMetrics) that will replace it, with paired “net (assumed)” and “net (measured)—pending” columns already in place. Strategies with two-legged turnover are known casualties in waiting: a market-neutral wedge long–short variant, now reported separately with the non-model exhibits, already died under the placeholder assumption (68%/yr gross to –12% net), and measured spreads can only be less kind.

Monthly tenors only. Bloomberg surfaces provide monthly expiries; the weekly-tenor segment—where crash insurance demand is most acute—is unobserved. Weekly-tenor replication now travels with the separately reported non-model VRP-harvest exhibits (Appendix A.6).

No fill simulation. Executions are assumed at model prices net of assumed costs; there is no order-book or fill simulation, and defined-risk digital/put-spread structures are approximated.

Survivorship handling is pending. The CRSP delisting-return merge has not yet been applied (Section 3.3); left-tail economics for exiting names may be understated. Pre-registered as Appendix A.6.

Universe growth. Coverage grows from ~ 100 to ~ 500 names over the sample; early-sample results are estimated on a thinner, larger-cap universe, and cohort analysis (reported separately with the non-model exhibits) suggests composition matters.

Calibration statistics are unconditional. Coverage is reported unconditionally by horizon and level; the conditional-coverage, PIT, expected-shortfall, and forecast-comparison tests are specified but pending (Section 5.4).

The conformal stage is pending. GRAFT-Q’s cross-sectional conformal layer (Section 4.5) is specified and pre-registered but not yet reflected in reported numbers.

10 Conclusion and future work

A single horizon-conditioned implicit quantile network can deliver essentially exactly calibrated conditional return distributions for hundreds of single names at every horizon from one week to three months—everywhere except the region where price-based features are informationally blind. There, the honest fix is a two-parameter EVT graft estimated on the past, not a larger network. The calibrated distributions turn the variance-risk-premium literature conditional: crash insurance is overpriced even when the buyer is right about the risk, which relocates the puzzle from forecasting to risk-bearing capacity, consistent with demand-based option pricing and probability-weighted demand.

The pre-registered extensions are: GRAFT-Q’s cross-sectional conformal stage and the evaluation battery of Section 5.4; measured OptionMetrics spreads replacing the placeholder cost assumption (Appendix A.5); CRSP delisting-return tails; the staged days-to-earnings feature as

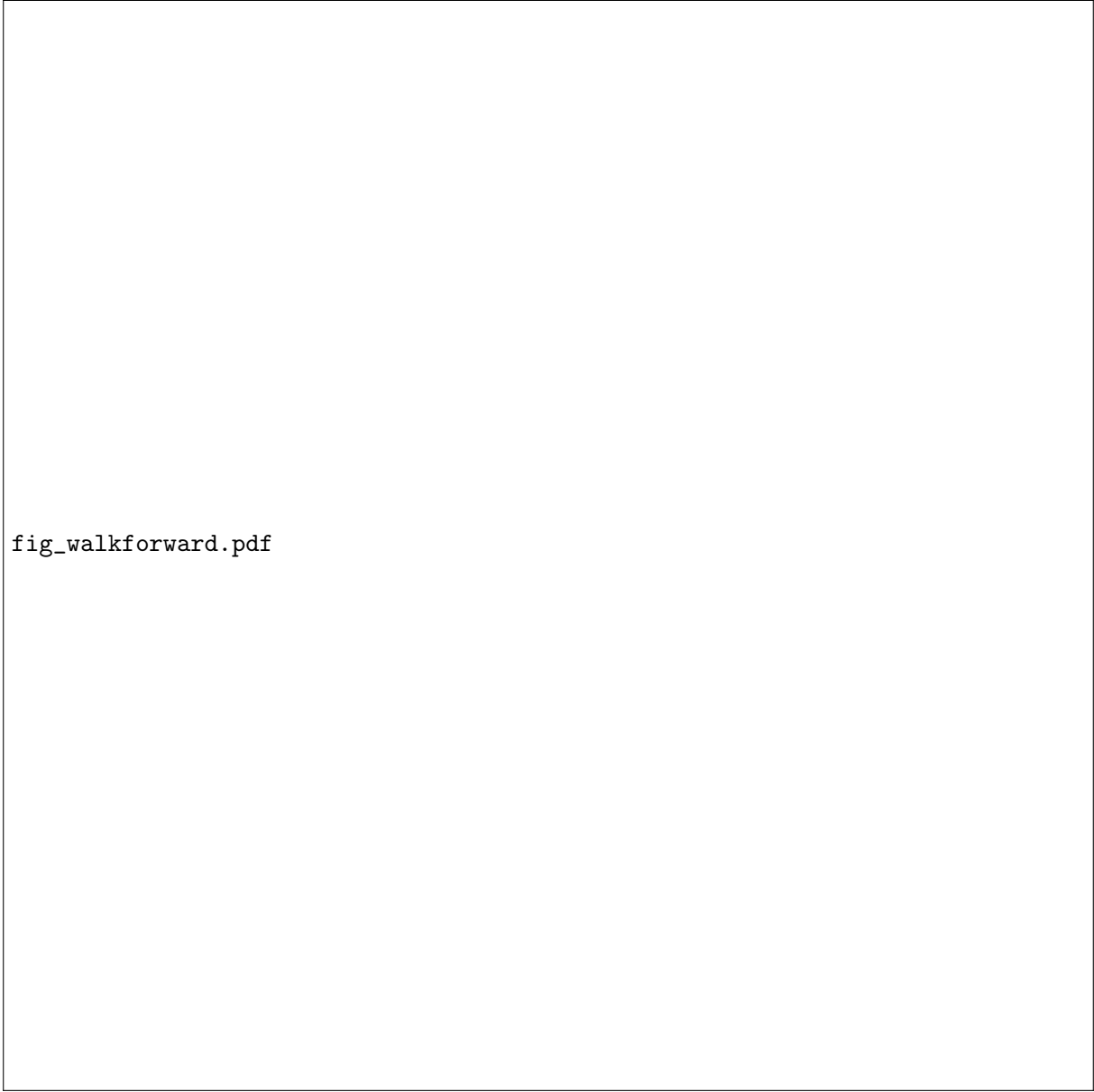
the first direct attack on the idiosyncratic-event ceiling; dispersion structures across the single-name/index wedge; and dealer-positioning (GEX-type) features for the supply side of Section 8.2. The July-2026 live holdout will be reported in a revision regardless of outcome.

References



fig_architecture.pdf

Figure 1: Model architecture. State features, horizon, and a sampled quantile level (cosine-embedded) feed a single implicit quantile network whose output, averaged over a $K = 3$ ensemble, forms the neural quantile body; a walk-forward generalized Pareto tail, estimated strictly on past data, is spliced below the model's own 5% quantile.



fig_walkforward.pdf

Figure 2: Walk-forward protocol. Each year's predictions come from a model trained only on data whose labels end 95 days earlier; the embargo exceeds the longest prediction horizon plus slack, so no training label overlaps any prediction window.

fig_calibration.pdf

Figure 3: Out-of-sample calibration. (a) Body and upper tail: empirical coverage (bars; ranges across horizons as error bars) against nominal levels (red rules). (b) Left tail: the 5% quantile's breach rate for the raw IQN, the heavy- τ -sampling variant, and the EVT splice, against the nominal 5% line.

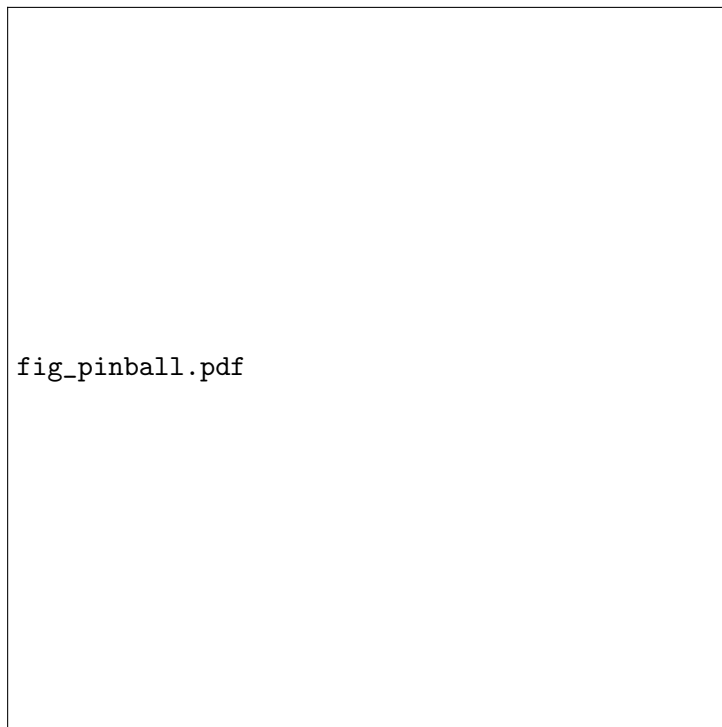
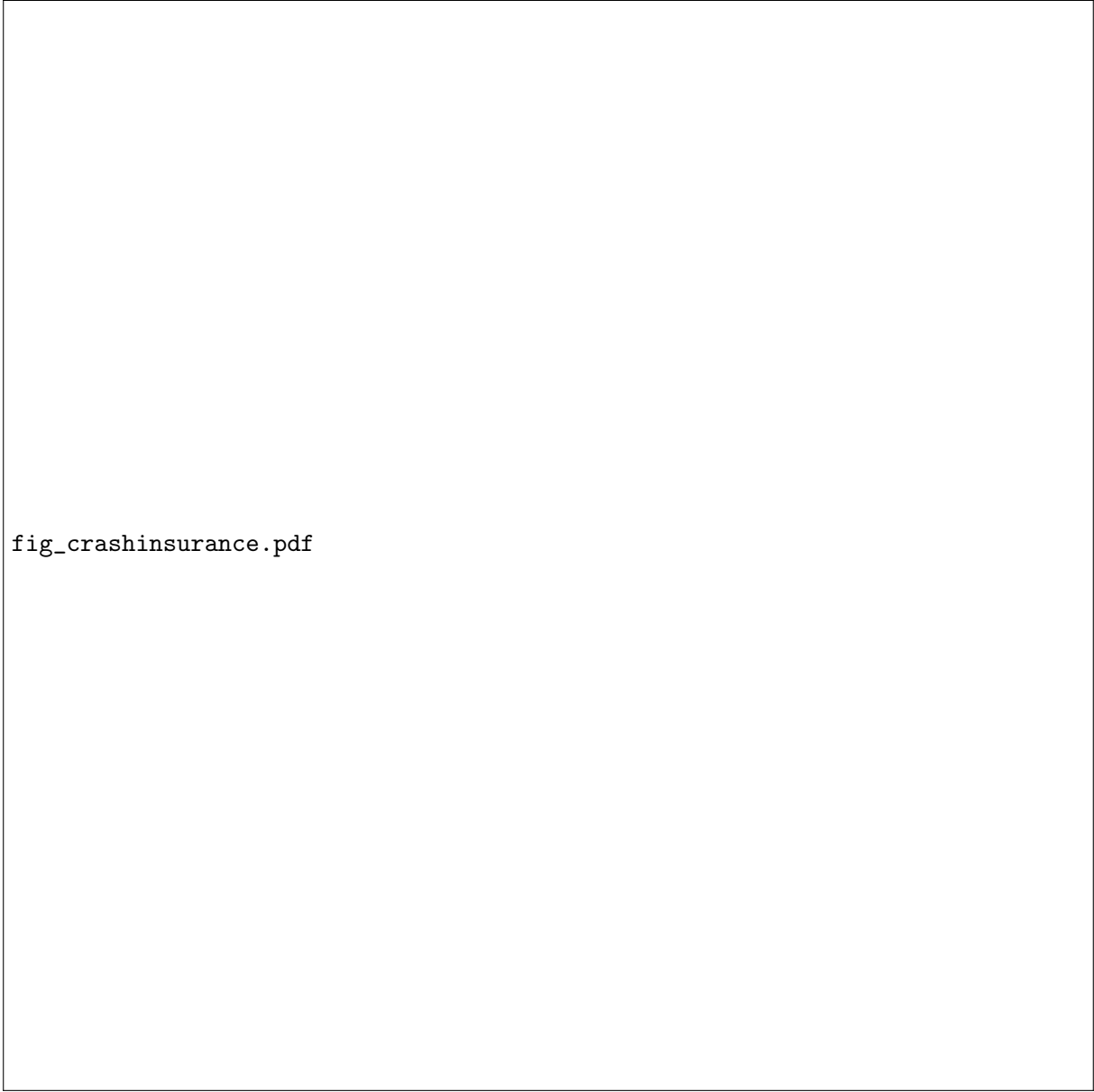


Figure 4: Dimension-reduction tournament. Mean walk-forward pinball loss for raw 17 features, PCA-6, and block-PCA-6. Note the compressed axis: differences are in the fourth decimal.



fig_crashinsurance.pdf

Figure 5: The price of crash insurance. (a) Months flagged as elevated-tail by the IQN's distributional trigger versus a GJR-GARCH- t trigger. (b) Mean monthly net return of buying 5%-physical-quantile puts conditional on the IQN trigger.

A Trading system and full results

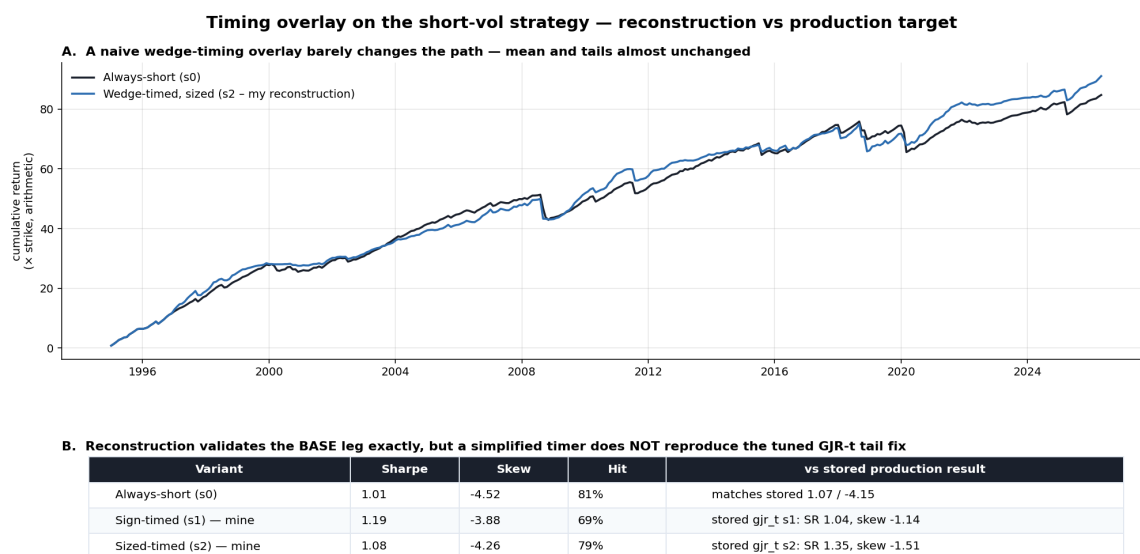
One-line takeaway: the model’s economic value shows up in the tail, not the mean—model-driven timing and sizing turn a crash-exposed short-volatility premium into a book whose worst episodes are survivable, and the exhibits below document that transformation. This appendix contains the economic application, per the design principle that the paper’s contribution is the calibrated distribution and its implications; the trading system is the measurement apparatus. All results are walk-forward with purge and embargo (models are retrained annually and scored only on data after each training cutoff, with a 95-day gap so that no training label overlaps any evaluation or trading window; Section 5). **All “net” figures in this appendix use the placeholder cost assumption of 10% of premium.** Each cost-dependent exhibit carries a paired column for measured OptionMetrics spreads, currently pending (Appendix A.5). Following this draft’s scope revision, the appendix reports only strategies in which the IQN model’s forecasts enter the signal, timing, or sizing; pure variance-risk-premium harvest strategies without model input, previously reported here, are removed to keep the paper’s scope to model-driven applications and are reported separately.

A.1 Signal construction and the $\mathbb{P}$ – $\mathbb{Q}$ wedge

The core signal compares the smile-implied ($\mathbb{Q}$: the risk-neutral measure the option surface embeds) 10th-percentile strike with the model’s physical ($\mathbb{P}$: the measure under which returns are actually realized) breach probability for that strike. Strikes are set at the risk-neutral 10th percentile; when the calibrated physical breach probability is materially lower, the strike is rich, and the system sells defined-risk structures (digitals or put spreads) at that strike monthly. Strike depth is modulated by a validated model-agreement (“conflict”) signal between the GJR-GARCH- t (21d) benchmark and the IQN: agreement deepens strikes, conflict shallows them. Figure 6 audits this signal chain on the index short-variance leg: the reconstructed base leg matches the stored production series, while a naive wedge z -score overlay barely changes the path—the documented tail improvement requires the tuned model-timing specification, not a generic wedge threshold.

A.2 Strategy variants

Table 4 collects the model-based variants retained in this draft. (The switcher’s rule in one sentence: position depth and size follow the model-agreement signal of Appendix A.1—when the GJR-GARCH- t and IQN distributions agree, the book sells deeper defined-risk strikes at full size; when they conflict, it shallows strikes and cuts size, and in stress the timing leg takes the book net long volatility.) The GARCH/IQN regime switcher attains $SR \approx 1.1$ – 1.16 with the best crisis profile of any variant tested: it goes net long volatility in stress, and its timing leg recorded $+5.5$ in the GFC and $+10.1$ in the COVID crash (units as recorded in the experiment logs: multiples of the leg’s monthly premium base). Non-model variants—the `sell_10` selection book (plain VRP selection with no model input), the regime ladder, the index variance leg and



The stored -1.5 skew comes from the production GJR-t leverage timing that goes net-long-variance INTO crashes (e.g. pre-COVID) — a tuned signal+sizing spec that lives in the pipeline code, not captured by a naive wedge z-score.

Figure 6: Reconstruction audit of the timing layer on the index short-variance leg. Panel A plots cumulative return (multiples of strike, arithmetic) for the always-short leg (s0) and a naive wedge-timed, sized overlay (s2) built directly from the $\mathbb{P}$ - $\mathbb{Q}$ wedge z -score, 1996–2026: the overlay barely changes the path. Panel B compares summary statistics: the reconstructed base leg (Sharpe 1.01, skew -4.52) matches the stored production base leg (1.07, -4.15), validating the data chain end to end, but the naive timers (Sharpe 1.08–1.19, skew -3.9 to -4.3) do not reproduce the stored production GJR- t timing (Sharpe 1.35, skew -1.51), whose tuned signal-plus-sizing rule goes net long variance into crashes. For risk-taking the exhibit carries two messages: the base data and P&L chain are independently reproducible, and the switcher’s crisis profile is a property of the specific frozen production rule rather than of any generic wedge-threshold overlay.

its 50/50 combination, the master book aggregation, the market-neutral wedge long–short, and long cheap OTM calls—are reported separately with the other pure VRP-harvest exhibits.

Three exhibits document the model-driven legs. Figure 7 isolates the timing mechanism on the index short-variance leg: model-timed variants earn a similar Sharpe to the always-short leg but with materially less negative skew—timing adds tails, not mean. Figure 8 shows that the timing model itself is stable across training windows, and, just as importantly, where it is blind. Figure 9 shows the switcher book’s operating profile on the single-name universe: cumulative net return, rolling win rate, trade volume, and compounded wealth.

A.3 Capital allocation: the committed-fraction frontier

Returns on committed capital overstate wealth dynamics unless the committed fraction f of wealth is specified. Table 5 reports 21-year compounded outcomes by f for the switcher-style book (retained here because the underlying book is the model-based GARCH $\times$ IQN switcher):

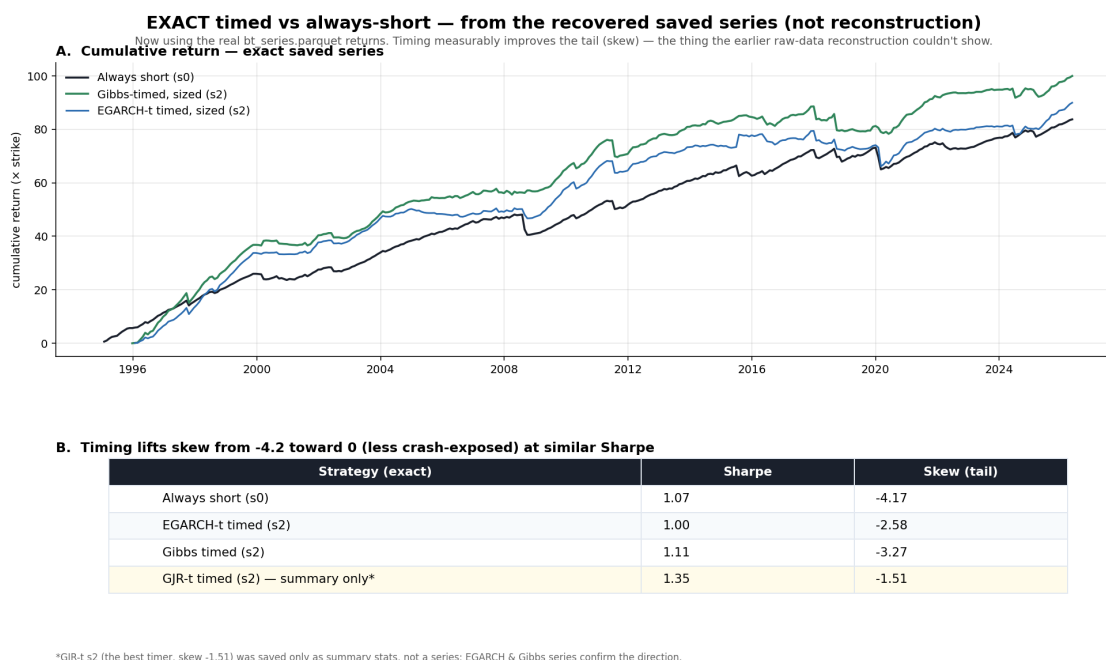


Figure 7: Model timing on the index short-variance leg, from the exact stored backtest series. Panel A plots cumulative return (multiples of strike) for the always-short leg (s0) and two conditional-volatility-timed, sized variants (Gibbs-timed and EGARCH- t -timed, s2), 1996–2026; the timed paths sit above the always-short path for most of the sample. Panel B reports the statistic timing is for: monthly-return skewness improves from -4.17 (always short) to -2.58 (EGARCH- t) and -3.27 (Gibbs) at similar Sharpe (1.07 versus 1.00–1.11), with the best timer (GJR- t , saved as summary statistics only) at Sharpe 1.35 and skew -1.51 . The mechanism is the leverage effect: conditional-volatility models raise forecast volatility quickly enough in stress for the rule to cut or reverse short-variance exposure, converting an unconditionally crash-skewed premium into a less crash-exposed one. That matters for risk-taking because the short-volatility book’s existential risk is the bunched left tail, not the average month.

full commitment produces spectacular CAGR (135%) with intolerable path risk (worst month -28% of wealth; max drawdown -29%), while $f = 25\%$ yields 27% CAGR with a -7.7% drawdown. A \$10M mandate illustration (switcher, $f = 25\%$): mean $+\$66\text{k}/\text{month}$, worst month $-\$1.64\text{M}$, wealth max drawdown -17% , CAGR 7.9%. Figure 9 (bottom-right panel) shows the corresponding compounded wealth path for the switcher at $f = 25\%$, whose terminal multiple of roughly $4.7\times$ is in line with this illustration.

A.4 Negative results catalog

Half the project’s information is in what failed. Table 6 catalogs the model-associated negative result, economically motivated ex ante and reported with the same walk-forward discipline as the positive results. A complementary caution from the timing side is visible in Figure 8: both GJR- t fits forecast calm on the eve of the February 2020 and April 2025 shocks, so even a parameter-stable timing model cannot trigger insurance ahead of an event that has not yet

Table 4: Model-based strategy variants retained in this draft: net performance under placeholder costs, with pre-registered measured-cost column. Monthly rebalancing, 2005–2026 walk-forward, ~ 543 -name universe. Non-model VRP-harvest variants previously tabulated here are reported separately.

Variant	Net ann. return ^c	Net SR (assumed 10%)	Net SR (measured) ^a	Win rate (monthly)	Notes
GARCH $\times$ IQN switcher	—	1.1–1.16	<i>pending</i>	—	best crisis profile ^b

^a Pre-registered: to be restated with measured OptionMetrics effective spreads (Appendix A.5).

^b Goes net long vol in stress; timing leg +5.5 (GFC), +10.1 (COVID), in experiment-log units.

^c On committed capital, net of the assumed 10%-of-premium cost.

Dashes: not recorded in the experiment logs used for this draft.

Table 5: Committed-fraction (Kelly-style) table: 21-year compounded outcomes by fraction f of wealth committed, switcher-style book.

	$f = 100\%$	$f = 50\%$	$f = 25\%$	$f = 10\%$
CAGR	135%	58%	27%	10%
Worst month (% of wealth)	–28%	–14%	–7%	–2.8%
Max drawdown	–29%	–15%	–7.7%	–3.1%
<i>\$10M illustration (switcher, $f = 25\%$): mean +\$66k/mo, worst –\$1.64M, wealth maxDD –17%, CAGR 7.9%.</i>				

moved returns (Section 7)—which is why the retained designs route tail risk through defined-risk structure and sizing rather than through forecast-triggered protection. Six further dead ends concerning the non-model VRP-harvest books (vol-targeted sizing, a short-index-futures overlay, skew-richness sorting, per-name risk parity, post-crash doubling, and selling the steepest smiles) are reported separately with those exhibits.

A.5 Execution costs: from assumption to measurement

Every net figure above assumes costs of 10% of premium—prominent, deliberate, and temporary. The pre-registered replacement is a measured cost model estimated from OptionMetrics quotes with the following design, fixed in advance of seeing the data: the effective relative half-spread is modeled as a function of (i) moneyness K/S (spreads widen into the wings), (ii) tenor (shorter tenors trade wider in relative terms), (iii) market-capitalization bucket, and (iv) the VIX regime (spreads widen when the book’s positions are largest in absolute premium). The fitted surface will be applied trade-by-trade in the walk-forward, producing the “net (measured)” column already reserved in Table 4. Figure 10 reserves the corresponding exhibit. The cost hypotheses previously registered for the non-model variants travel with those exhibits to the separate report.

A.6 Pre-registered robustness extensions

(a) *Delisting-return tails.* The CRSP delisting-return merge will append terminal returns for names exiting the panel. Registered prediction: calibration statistics change negligibly (delisting

Table 6: Negative results involving the model’s forecasts. The row was a plausible design ex ante; it is reported with the same walk-forward discipline as the positive results.

Design	Outcome	Diagnosis
(1) Model-timed crash insurance (buy 5% puts on IQN fat-tail trigger; 525 trigger months vs. 6 for GARCH)	$-1.7\%/mo$ ($t = -1.9$)	Insurance overpriced even conditional on correct risk forecasts (Sec. 8)

months are a small fraction of observations), while single-name short-put economics deteriorate modestly. Exhibits to be restated are listed in Appendix C.

(b) *Weekly-tenor replication.* The weekly-tenor replication pre-registered in the first draft concerns the non-model VRP-harvest book and is now reported separately with those exhibits.

B Implementation details

Panel construction. Approximately 543 names, 2005–2026, $\sim 450,000$ row $\times$ horizon samples; horizons of one week to three months on business-day grids; entries settle over $t+1$ through $t+h$.

Network and training. Horizon-conditioned IQN (Section 4) with cosine τ -embedding merged multiplicatively into the state trunk; $K = 3$ ensemble over initialization and data-order seeds; pinball loss with uniform τ -sampling (v2c’s left-weighted sampler evaluated and rejected, Section 6.2). All hyperparameter and architecture selection used nested walk-forward exclusively: an inner walk-forward within each training span selects configurations; the outer span is touched once.

Features. Base nine: realized-vol block (cascaded lagged windows), leverage, own-IV level and skew, VIX level, vol term slope, futures basis. Extended seventeen adds VIX9D/VIX, 10y yield, 2s10s, HY and IG OAS, trailing-63d market return, drawdown-from-peak, distance-from-200dma. Standardization on lagged windows only; macro NaNs properly imputed (Section 6.4); regime indicators are expanding-percentile and point-in-time.

EVT splice. Threshold at the body’s 5% quantile in model probability coordinates; GPD fit by maximum likelihood to past standardized exceedances only, refreshed with each annual retrain; splice continuous at the threshold; body above $\tau = 0.05$ untouched.

Baselines. GJR-GARCH- t on 21-day windows (?) for wedge timing and the conflict signal.

Anti-overfit instruments. Annual retraining; 95-day purge/embargo (exceeds longest horizon plus slack); frozen strategy rules; fresh-universe replication on 496 independently assembled names; live post-July-2026 holdout; nested walk-forward for all selection.

C Pre-registered data extensions (WRDS)

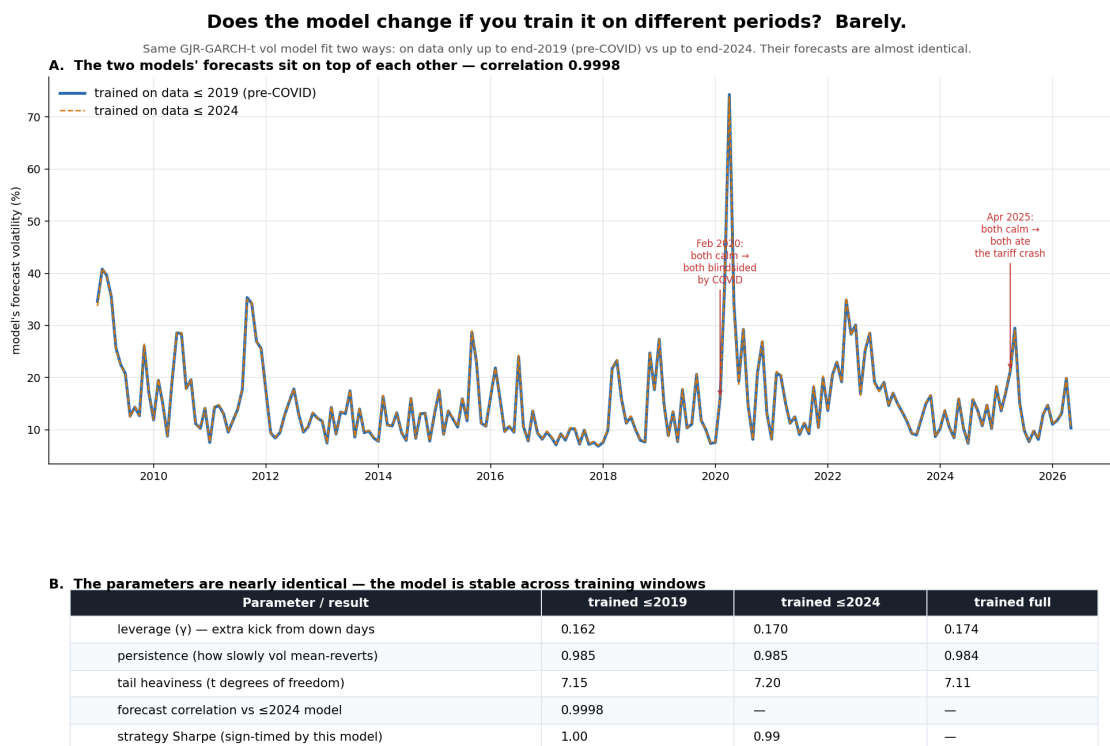
Table 7 maps each pending WRDS dataset to what it replaces and the exhibits it will restate. The paper’s structure is designed so these are restatements, not restructurings: affected tables

carry paired assumed/measured columns, and affected figures have reserved placeholders.

Table 7: Pre-registered WRDS data extensions: dataset, role, and exhibits to be restated.

WRDS dataset		Replaces / adds	Exhibits restated
OptionMetrics	IvyDB	Placeholder 10%-of-premium cost assumption → measured effective-spread surface (App. A.5)	Table 4 (measured column); Figure 10; App. A.3 wealth outcomes
CRSP delisting returns		Current partial survivorship handling → terminal-return tails for exiting names (App. A.6a)	Table 3 (verification only); Table 4
CRSP names/events		Point-in-time universe membership audit	Section 3.3; App. B panel counts

Registered before data access; hypotheses for each extension are stated in Appendices A.5 and A.6.



Why: GARCH is a CONDITIONAL model — it re-reads recent returns to forecast, so the training cutoff mostly re-estimates the same stable parameters. Reassuring for overfitting. BUT identical ≠ safe: both models forecast CALM entering COVID and April 2025, because no vol model can predict a shock that hasn't shown up in returns yet.

Figure 8: Stability diagnostic for the timing model. The same GJR-GARCH- t volatility model is fit on data through 2019 (pre-COVID) and through 2024; panel A overlays their forecast-volatility paths (correlation 0.9998), and panel B shows near-identical parameters (leverage 0.162 versus 0.170, persistence 0.985, tail degrees of freedom 7.15 versus 7.20) and near-identical sign-timed strategy Sharpe (1.00 versus 0.99). Because GARCH is a conditional model that re-reads recent returns, moving the training cutoff mostly re-estimates the same stable parameters—evidence that the timing leg's behavior is not an artifact of a particular estimation window. The annotated episodes cut the other way: both fits forecast calm entering February 2020 and April 2025, the timing-model face of the informational ceiling of Section 7—no returns-conditioned model can foresee a shock that has not yet appeared in returns. Stability supports trusting the rule; the blind spots explain why defined-risk structure and sizing, not forecasts alone, must carry the tail.

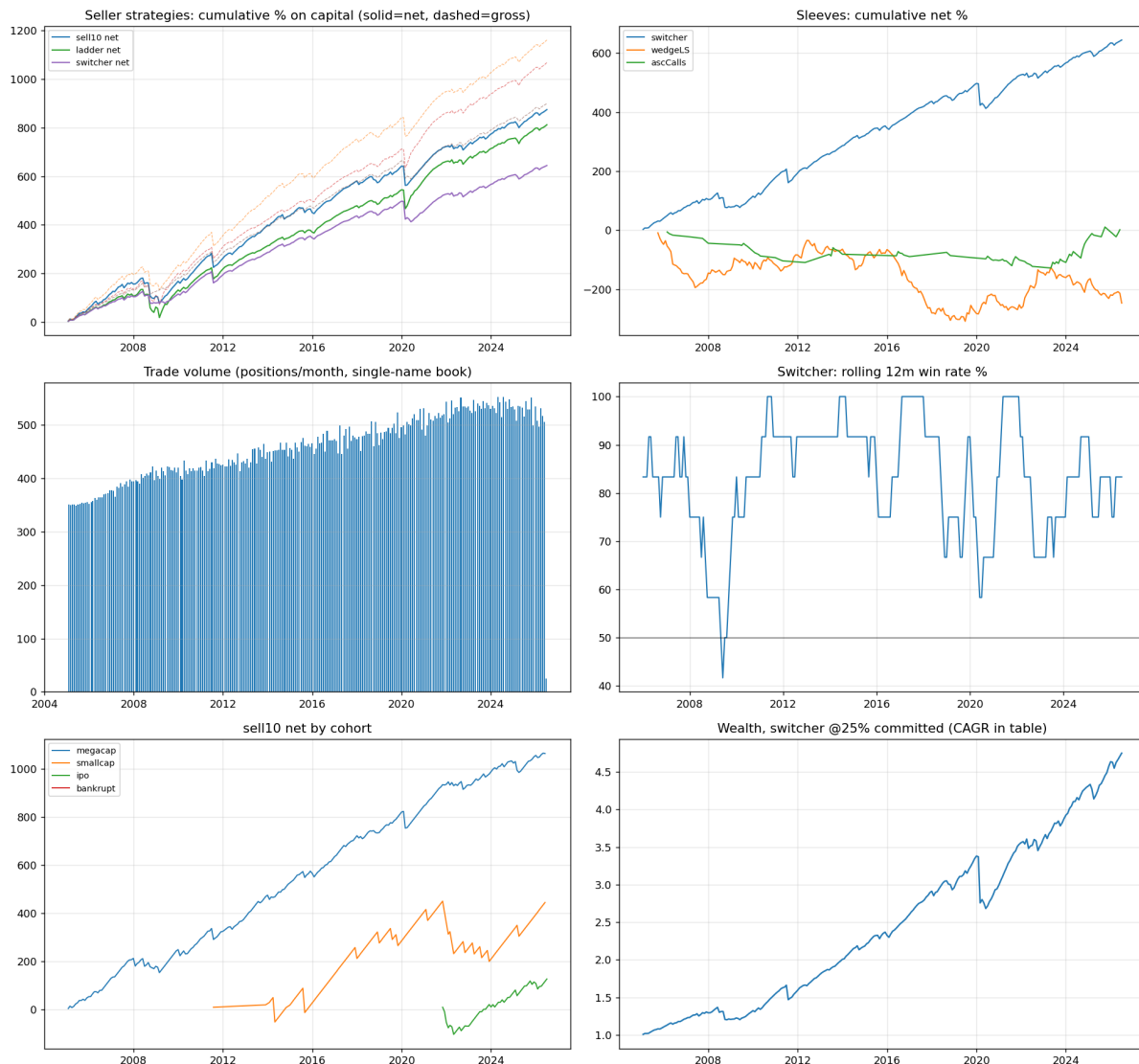


Figure 9: Operating profile of the model-driven switcher book on the single-name walk-forward universe. Model-based panels are the focus; the non-model series (`sell10`, ladder, and the cohort decomposition), which are reported separately, appear for context only. Top row: cumulative net percent on capital for the switcher next to the two non-model selection books (left), and the model-based sleeves in isolation (right), where the switcher sleeve compounds to roughly +640% while the wedge long-short and ascension-call sleeves oscillate near or below zero. Middle row: trade volume of the single-name book, roughly 350 rising to 550 option positions per month as options coverage broadens (left), and the switcher’s rolling 12-month win rate (right), which lives at 70–100% but dips to roughly 42% in 2009 and below 60% in 2020—wins are steady, losses bunch in crash clusters. Bottom row: the `sell10` cohort decomposition (left; context only) and compounded wealth of the switcher at 25% committed capital (right), ending near $4.7\times$, in line with the $f=25\%$ illustration of Appendix A.3; together the panels give the sizing-relevant facts—the edge is realized across thousands of small positions per year, and the failure mode is concentrated episodes rather than chronic bleed.

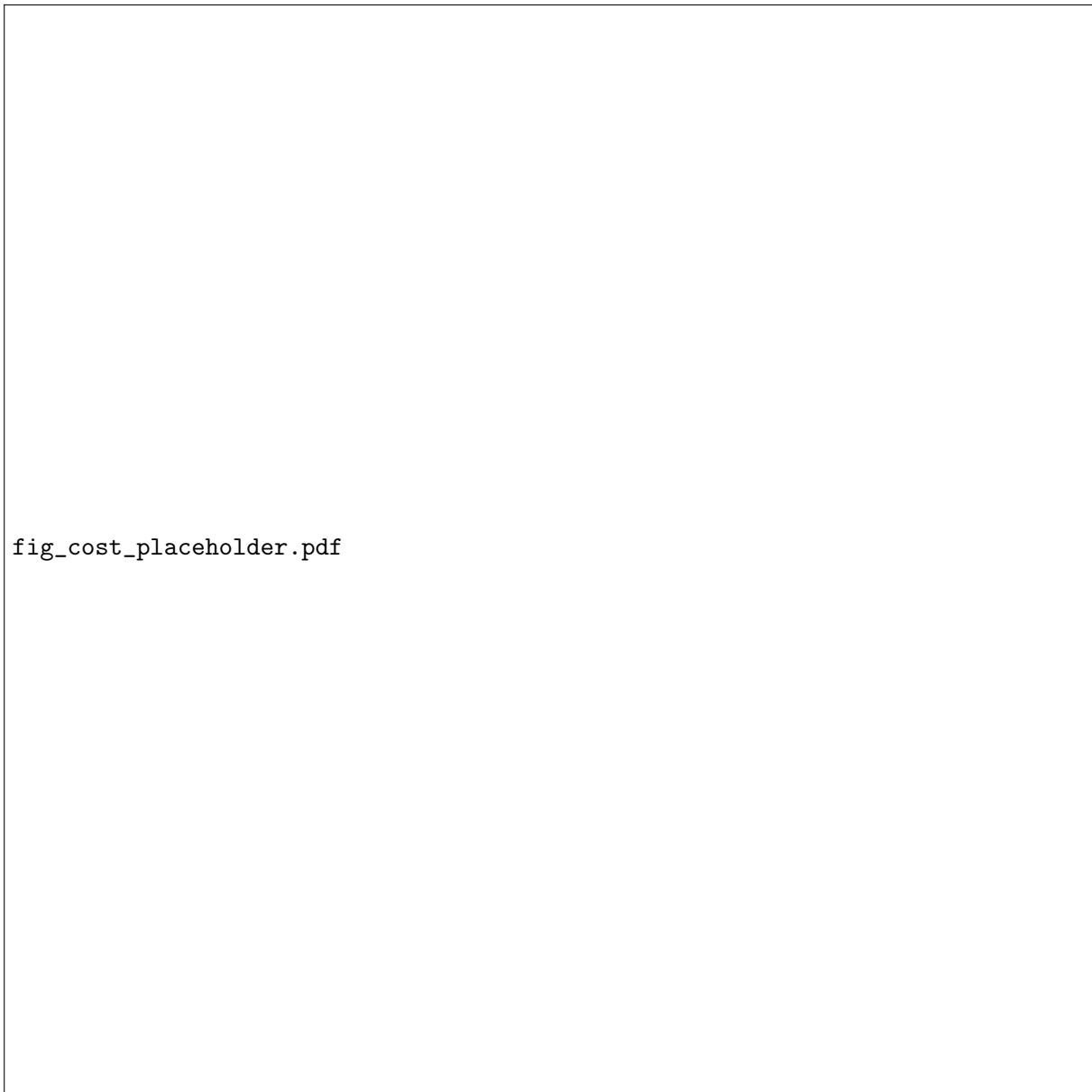


Figure 10: Placeholder (pre-registered exhibit). The measured effective-spread surface from Option-Metrics, to be overlaid with the constant 10%-of-premium assumption used throughout this draft.